

USE CASE-2

EBOOK SUBSCRIBER INSIGHTS

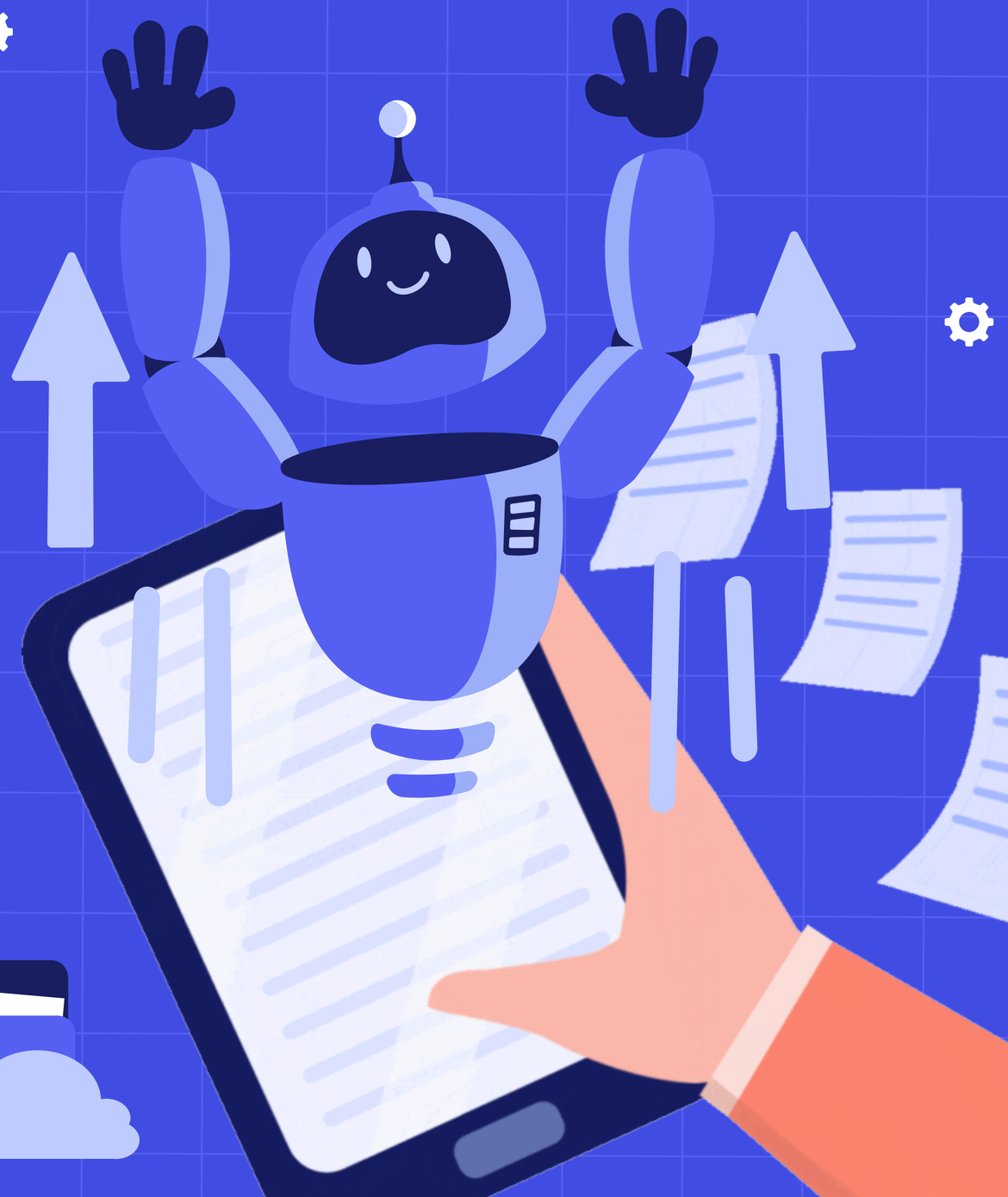
USING EDA

PROBLEM STATEMENT OVERVIEW

- Digital subscription platforms face persistent churn that impacts profitability
- Cannot identify which signals distinguish high-engagement from low-engagement subscribers
- Lack of systematic methods to segment users for targeted retention
- Generic retention strategies fail because churn drivers vary by segment

Research Question:

- What behavioral and demographic signals distinguish high-engagement subscribers from low-engagement ones?









LITERATURE REVIEW

Previous studies on digital subscription platforms highlight the importance of understanding subscriber engagement and churn. Research shows that user activity patterns and demographic characteristics can help distinguish different levels of engagement and identify potential churn-risk users. EDA provides a useful way to explore these patterns and understand subscriber behavior.

THEME 1 -ENGAGEMENT DRIVERS

- **Early engagement (first 30 days)**

Strong early activity can indicate continued interest and improve the likelihood of subscriber retention.

- **Content diversity and feature breadth**

Exploring different content and platform features is associated with deeper engagement and potentially lower churn.

- **User experience friction**

Difficult navigation, complicated interfaces, or barriers to content access can reduce engagement and encourage users to disengage.

- **Platform usability and content quality**

Both relevant, high-quality content and a smooth, user-friendly platform experience are important for maintaining subscriber engagement.







THEME 2 – CHURN SIGNALS

- **Declining login frequency**

A gradual decrease in login activity can indicate weakening interest and may serve as an early warning sign of potential churn.

- **Reduced session duration and feature usage**

Shorter reading sessions and reduced interaction with platform features can reflect declining subscriber engagement before churn occurs.

- **Lower interaction with new releases**

Reduced engagement with newly added books or content may signal that subscribers are becoming less interested or less active on the platform.

- **Churn drivers differ across segments**

Different subscriber groups may churn for different reasons, such as price sensitivity, limited content, or poor platform experience, making segment-specific analysis important.





THEME–3: SUBSCRIBER SEGMENTATION

- **Super Users** — Show consistently high engagement through frequent logins, longer sessions, and regular content interaction, making them the lowest-risk group for churn.
- **Casual Users** — Display moderate engagement and occasional platform activity, placing them at a moderate level of churn risk.
- **Inactive Users** — Show minimal interaction with the platform and limited content usage, indicating a higher likelihood of disengagement and churn.
- **At-Risk Users** — Show a noticeable decline in engagement over time, making them a high-priority group for early retention and re-engagement efforts.



THEME–4: EDA FOR CHURN PREDICTION

- **Univariate analysis reveals engagement distributions —** Examining individual variables helps identify common engagement patterns, usage levels, and differences among subscribers.
- **Visualizations identify churn inflection points —** Charts and graphs help highlight changes in subscriber activity that may indicate increasing churn risk.
- **Cohort retention curves show segment-specific trajectories —** Retention trends across different subscriber groups reveal how engagement and churn patterns change over time.
- **User journey maps pinpoint intervention opportunities —** Mapping subscriber interactions helps identify stages where engagement drops and where targeted retention actions can be introduced.



THEME–5: PERSONALIZATION & RETENTION

- **Personalized recommendations** — Tailored content recommendations can improve subscriber relevance and engagement compared with generic content suggestions.
- **Content freshness and release cadence** — The availability of new and regularly updated content can influence subscriber engagement and may help identify potential churn patterns.
- **Relevant communication** — Timely and relevant messages can encourage inactive subscribers to return and increase re-engagement.
- **Segment-specific strategies** — Adapting retention strategies to the needs and behavior of each subscriber segment can improve the effectiveness of engagement efforts.







METHODOLOGY AND DESIGN

Analytical Approach

Data Collection & Preprocessing

- eBook subscriber dataset with demographics and engagement metrics

Subscriber Segmentation

- Active, Inactive, Churned classification

Engagement Scoring Model

- Weighted formula combining key metrics







METHODOLOGY AND DESIGN

Data Collection & Preprocessing

- **eBook subscriber dataset with demographics and engagement metrics —** The dataset contains subscriber-level information such as demographic characteristics, login frequency, session duration, reading activity, content interaction, and other relevant engagement indicators.
- **Data cleaning and preparation —** Missing values, duplicate records, and inconsistent entries are identified and handled to ensure the dataset is suitable for analysis.
- **Feature preparation —** Relevant demographic and behavioral variables are selected and transformed into a consistent format for further analysis.







METHODOLOGY AND DESIGN

Subscriber Segmentation

- **Active, Inactive, and Churned classification** — Subscribers are categorized based on their level of platform activity and subscription status.
- **Behavior-based grouping** — Engagement patterns such as login frequency, reading activity, session duration, and feature usage are analyzed to distinguish different subscriber groups.
- **Risk identification** — Segmentation helps identify highly engaged users as well as inactive and potentially at-risk subscribers.







METHODOLOGY AND DESIGN

Engagement Scoring Model

- **Weighted formula combining key metrics —** An engagement score is calculated by assigning different weights to important behavioral indicators such as login frequency, session duration, reading activity, and content interaction.
- **Overall engagement measurement —** The combined score provides a single measure of each subscriber's engagement level.
- **Ranking and comparison —** Subscribers can be ranked and grouped into low-, medium-, and high-engagement categories, making it easier to compare their churn risk and behavioral patterns.



VISUALIZATION AND ANALYSIS

- **Pie Charts: Segment Distribution** — Show the proportion of Active, Inactive, and Churned subscribers for an overall view of the user base.
- **Bar Charts:** Genre & Behavioral Patterns — Compare preferred eBook genres and key behaviors such as reading frequency, logins, and content interaction.
- **Heatmaps: Correlation Analysis** — Highlight relationships between engagement metrics and help identify factors associated with higher or lower churn.
- **Retention Curves:** Cohort Churn Trajectories — Track retention over time across different subscriber cohorts and identify segments with faster engagement decline or churn.



EXPECTED FINDINGS

- **Clear subscriber segmentation** — Group users into Super, Casual, Inactive, and At-Risk segments based on engagement.
- **Key churn factors** — Identify behavioral patterns such as declining logins, shorter sessions, and reduced content interaction.
- **Early warning signals** — Detect declining engagement early to identify subscribers who may be likely to churn.
- **Segment-specific strategies** — Recommend targeted retention actions based on the behavior and needs of each subscriber group.



CONCLUSION

- **Churn follows identifiable behavioral patterns** — Changes in login frequency, reading activity, session duration, and content interaction can indicate declining subscriber engagement.
- **EDA reveals engagement signals and intervention points** — Exploratory analysis helps uncover important behavioral trends and identify where timely action can prevent further disengagement.
- **Segment-specific strategies outperform generic approaches** — Understanding the unique behavior of each subscriber segment allows platforms to design more relevant and effective retention strategies.
- **Data-driven retention is essential for platform sustainability** — Using subscriber data to monitor engagement, detect churn risks, and personalize interventions can support stronger long-term subscriber retention.

THANK YOU

